Pure Mathematics 3 Practice Paper 2 (2026 style)

Answers and mark guide

M = method mark. A = accuracy mark; it needs the related M mark. B = independent mark. AG = the answer is given in the question. Accept other correct methods.

Question 1 **[4]**

$\ln\frac{x^2}{x-a} = \ln(4a)$ M1

$x^2 = 4a(x-a)$, so $x^2 - 4ax + 4a^2 = 0$ A1

$(x-2a)^2 = 0$ M1

$x = 2a$ only. This is a valid root, because $2a > a > 0$. A1

Question 2 **[5]**

(a) $(1+bx)^{-\frac{1}{2}} = 1 - \frac{1}{2}bx + \frac{3}{8}b^2x^2 + \ldots$ B1

Multiply by $1 + ax$: $1 + (a - \frac{1}{2}b)x + (\frac{3}{8}b^2 - \frac{1}{2}ab)x^2 + \ldots$ M1

Compare coefficients: $a - \frac{1}{2}b = 0$ and $\frac{3}{8}b^2 - \frac{1}{2}ab = 2$ A1

$a = \frac{1}{2}b$ gives $\frac{1}{8}b^2 = 2$, so $b = 4$ and $a = 2$ A1

(b) $|x| < \frac{1}{4}$ B1

Question 3 **[6]**

(a) $R = 5$ B1

$\tan\alpha = \frac{4}{3}$ M1

$\alpha = 0.9273$ A1

(b) The graph of $y = 5\cos(x-\alpha)$ starts at $y = 3$ when $x = 0$ and ends at $y = -3$ when $x = \pi$ B1

Between these, it rises to its greatest value, 5, at $x = \alpha$ B1

Two roots when $3 \leqslant k < 5$. ($k = 3$ gives $x = 0$ and $x = 2\alpha$. $k = 5$ gives one root only.) B1

Question 4 **[7]**

(a) $(x + \mathrm{i}y)^2 = -15 + 8\mathrm{i}$ gives $x^2 - y^2 = -15$ and $2xy = 8$ M1 A1

Eliminate y: $x^4 + 15x^2 - 16 = 0$, so $(x^2 + 16)(x^2 - 1) = 0$ and $x^2 = 1$ M1

The square roots are $1 + 4\mathrm{i}$ and $-1 - 4\mathrm{i}$ A1

(b) $z = \dfrac{(3 + 2\mathrm{i}) \pm \sqrt{(3 + 2\mathrm{i})^2 - 4(5 + \mathrm{i})}}{2}$ M1

The discriminant is $5 + 12\mathrm{i} - 20 - 4\mathrm{i} = -15 + 8\mathrm{i}$. Its square roots are $\pm(1 + 4\mathrm{i})$, from (a). A1

$z = 2 + 3\mathrm{i}$ and $z = 1 - \mathrm{i}$ A1

Question 5 **[9]**

(a) $\dfrac{\mathrm{d}}{\mathrm{d}x}(xy) = y + x\dfrac{\mathrm{d}y}{\mathrm{d}x}$ B1

$2x - y - x\dfrac{\mathrm{d}y}{\mathrm{d}x} + 2y\dfrac{\mathrm{d}y}{\mathrm{d}x} = 0$ M1

$(2y - x)\dfrac{\mathrm{d}y}{\mathrm{d}x} = y - 2x$, so $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{2x - y}{x - 2y}$ A1 AG

(b) Tangent parallel to the x-axis: $2x - y = 0$, so $y = 2x$. Then $x^2 - 2x^2 + 4x^2 = 12$. M1

$3x^2 = 12$, so the points are $(2,\ 4)$ and $(-2,\ -4)$ A1

(c) The tangent has the same gradient as the line: $\dfrac{2x - y}{x - 2y} = -1$ M1

$2x - y = 2y - x$, so $y = x$ A1

Substitute in the curve: $x^2 = 12$. As $k > 0$, the point of contact is $(2\sqrt{3},\ 2\sqrt{3})$. M1

$k = x + y = 4\sqrt{3}$ A1

Other method (same marks): put $y = k - x$ in the curve to get $3x^2 - 3kx + k^2 - 12 = 0$. A tangent gives equal roots, so $9k^2 - 12(k^2 - 12) = 0$ and $k^2 = 48$.

Question 6 **[8]**

(a) $\mathrm{f}(x) = \dfrac{A}{x + 1} + \dfrac{Bx + C}{x^2 + 1}$, so $4x = A(x^2 + 1) + (Bx + C)(x + 1)$ M1

$x = -1$ gives $A = -2$ A1

Compare coefficients: $B = 2$ and $C = 2$ A1 A1

$\mathrm{f}(x) = -\dfrac{2}{x + 1} + \dfrac{2x + 2}{x^2 + 1}$

(b) $\int \mathrm{f}(x)\,\mathrm{d}x = -2\ln(x + 1) + \ln(x^2 + 1) + 2\tan^{-1}x$ B1 B1 B1

$[\ldots]_0^1 = -2\ln 2 + \ln 2 + 2 \times \frac{1}{4}\pi - 0 = \frac{1}{2}\pi - \ln 2$ A1 AG

Question 7 **[10]**

(a) Volume $= \pi\int_0^a y^2\,\mathrm{d}x = \pi\int_0^a x\mathrm{e}^{-x}\,\mathrm{d}x$ B1

By parts: $\int x\mathrm{e}^{-x}\,\mathrm{d}x = -x\mathrm{e}^{-x} + \int\mathrm{e}^{-x}\,\mathrm{d}x$ M1

$= -x\mathrm{e}^{-x} - \mathrm{e}^{-x}$ A1

$\pi[-x\mathrm{e}^{-x} - \mathrm{e}^{-x}]_0^a = \pi(1 - (a+1)\mathrm{e}^{-a})$ A1 AG

(b) $1 - (a+1)\mathrm{e}^{-a} = \frac{1}{2}$, so $\mathrm{e}^a = 2(a+1)$ and $a = \ln(2a+2)$ B1 AG

(c) Let $\mathrm{h}(a) = \ln(2a+2) - a$: $\mathrm{h}(1.5) = 0.109$ and $\mathrm{h}(2) = -0.208$ M1

The sign changes and h is continuous, so the root is between 1.5 and 2 A1

(d) Use $a_{n+1} = \ln(2a_n + 2)$ correctly at least once M1

For example: 1.75, 1.7047, 1.6882, 1.6820, 1.6797, 1.6789, 1.6785, 1.6784. So $a = 1.68$. A1

Show enough iterations to be sure of 2 d.p., or show a sign change of h between 1.675 and 1.685 B1

Question 8 **[8]**

(a) A circle with centre $4 + 3\mathrm{i}$ B1

Radius 3. The circle touches the real axis at 4. B1

(b) Let C be the centre, so $OC = 5$. The greatest arg z is at T, where the tangent from O touches the circle. Angle $OTC = 90°$. M1

$OT = \sqrt{5^2 - 3^2} = 4$, so angle $COT = \tan^{-1}\frac{3}{4}$. This is also $\arg(4 + 3\mathrm{i})$. A1

Greatest arg $z = 2\tan^{-1}\frac{3}{4}$. Use $\tan 2\theta = \dfrac{2\tan\theta}{1 - \tan^2\theta}$. M1

$\tan 2\theta = \dfrac{3/2}{7/16} = \dfrac{24}{7}$, so the greatest value is $\tan^{-1}\frac{24}{7}$ A1

Short cut: the circle touches the real axis, so the other tangent from O is the real axis. The two tangents are symmetric about OC.

(c) $|z| = OT = 4$, with $\cos\theta = \frac{7}{25}$ and $\sin\theta = \frac{24}{25}$ M1

$z = \frac{28}{25} + \frac{96}{25}\mathrm{i}$ A1

Question 9 **[8]**

(a) $\overrightarrow{AP} = (2\lambda - 4)\mathbf{i} + (-\lambda - 1)\mathbf{j} + (2\lambda - 1)\mathbf{k}$ M1

$AP^2 = (2\lambda - 4)^2 + (\lambda + 1)^2 + (2\lambda - 1)^2 = 9\lambda^2 - 18\lambda + 18$ A1 AG

(b) $AP^2 = 9(\lambda - 1)^2 + 9$ M1

Shortest distance $= 3$ A1

Nearest point ($\lambda = 1$): $3\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ A1

(c) $9\lambda^2 - 18\lambda + 18 < 25$, so $9\lambda^2 - 18\lambda - 7 < 0$ M1

$(3\lambda + 1)(3\lambda - 7) < 0$, so $-\frac{1}{3} < \lambda < \frac{7}{3}$ A1

Length $= (\frac{7}{3} + \frac{1}{3}) \times |2\mathbf{i} - \mathbf{j} + 2\mathbf{k}| = \frac{8}{3} \times 3 = 8$ A1

Other method: by Pythagoras with the nearest point, half the segment is $\sqrt{5^2 - 3^2} = 4$.

Vectors are written with **i**, **j**, **k** here. Column vectors are equally correct.

Question 10 **[10]**

(a) $\frac{\mathrm{d}x}{\mathrm{d}t} = -kx$, where k is a positive constant B1

Separate and integrate: $\ln x = -kt + c$ M1 A1

$x = 100$ when $t = 0$, so $x = 100\,\mathrm{e}^{-kt}$ M1

$x = 40$ when $t = 6$, so $\mathrm{e}^{-6k} = 0.4$ and $x = 100\,(\mathrm{e}^{-6k})^{t/6} = 100 \times 0.4^{t/6}$ A1 AG

(b) In each 6-hour interval, the amount in the body is multiplied by 0.4, whatever the amount is at the start B1

Just after the nth dose: $100 + 100(0.4) + 100(0.4)^2 + \ldots + 100(0.4)^{n-1}$, a geometric progression M1

Sum $= \dfrac{100(1 - 0.4^n)}{1 - 0.4} = \dfrac{500}{3}(1 - 0.4^n)$ A1 AG

(c) Just after the nth dose, the amount is $\dfrac{D(1 - 0.4^n)}{0.6}$. This increases towards the sum to infinity, $\dfrac{D}{0.6}$, but never reaches it. M1

$\dfrac{D}{0.6} \leqslant 200$, so the greatest value of D is 120 A1

Where the grade separators are

A grade separator joins two topics, often in the last part of a question. The earlier parts of the same question are routine, so always do them first.

Main grade separators

Part	Topics joined	Hidden routine skill	First step
5(c)	Implicit differentiation + tangent lines (Pure 1)	A tangent has the same gradient as the line.	Put $\frac{\mathrm{d}y}{\mathrm{d}x} = -1$. Or put $y = k - x$ and use equal roots.
7(a)	Volume of revolution (Pure 1) + integration by parts	$V = \pi\int y^2\,\mathrm{d}x$	Square y first: $y^2 = x\mathrm{e}^{-x}$.
8(b)	Complex loci + circle tangents + $\tan 2A$	A tangent is perpendicular to the radius.	Draw O, the centre and the point of contact. Find the tangent length.
10(b), (c)	Differential equations + geometric series (Pure 1)	Every 6 hours, the amount is multiplied by 0.4.	Write the amounts after doses 1, 2 and 3. Look for the pattern.

Smaller steps of the same type

Part	Topics joined	Hidden routine skill	First step
3(b)	$R\cos(x - \alpha)$ + counting roots on a graph	Sketch $y = 5\cos(x - \alpha)$ for $0 \leqslant x \leqslant \pi$.	Find y at $x = 0$, $x = \alpha$ and $x = \pi$.
4(b)	Complex square roots + quadratic formula	The discriminant is the number in (a).	Write down $b^2 - 4ac$.
9(c)	Vectors + quadratic inequalities (Pure 1)	Solve $AP^2 < 25$.	Use the expression from (a).

Method for any new-style part

1. Find the routine skill inside the question.
2. Change each condition into an equation or an inequality.
3. Use the part before. "Hence" tells you that the earlier result is the key.
4. Work with letters, such as a, k and D, exactly as you work with numbers.
5. Always write the first step of a long part. The first marks are for method.